



17.5/20

Habib University

EE-371/CS-330 Computer Architecture – Fall 2025

Instructor: Dr. Shehzad Hasan

Time = 60 minutes

Quiz 02

Max Points: 20

Instructions:

- Smart watches, laptops, and similar electronics are strictly NOT allowed.
- Answer sheets should contain all steps, working, explanations, and assumptions.
- Attempt the quiz on clean papers with black/blue ink.
- Print your name and HU ID on all sheets.
- Turn in your question paper along with your answer sheets.
- You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	<i>Explain</i> the role of ISA in modern processors and instruction encodings and assembly language programming	
CLO 2	<i>Explain</i> the architecture and working of a single cycle processor	✓
CLO 3	<i>Design</i> the architecture to mitigate issues of a pipelined processor	
CLO 4	<i>Analyze the</i> performance of cache operations	

Question 1 [5 points]:

Compile the following C expression into RISC-V assembly language. Please archive all the necessary registers onto stack.

```
long long int summation (long long int elements[], size_t n) {
    size_t i = 0;
    long long int sum = 0;
    while (i < n) {
        sum = sum + elements[i];
        i++;
    }
    return sum;
}
```

Summation:

```
addi sp, sp, -48
sd x13, 40(sp) //address of array = 0x100
? { sd x15, 32(sp) //sum = 0
  sd x3, 24(sp) //i = 0
  sd x4, 16(sp) delete
  ld x5, 32(sp) }?
  ld x6, 24(sp) }?
loop: add x5, x5, x6(x13)
      addi x6, x6, 8
      blt x5, x6, loop
exit: add sp, sp, 48
```

2.5

X

Question 2 [5 points]:

Use a table similar to what is shown below, calculate the product of 5-bit unsigned integers 12 (01100) and 20 (10100). You should show the contents of each register on each step.

Iteration	Step	Multiplier	Multiplicand	Product
0	Initial values	0011	0000 0010	0000 0000
1	1a: 1 $\Rightarrow$ Prod = Prod + M Cand	0011	0000 0010	0000 0010
	2: Shift left Multiplicand	0011	0000 0100	0000 0010
	3: Shift right Multiplier	0001	0000 0100	0000 0010
2	1a: 1 $\Rightarrow$ Prod = Prod + M Cand	0001	0000 0100	0000 0110
	2: Shift left Multiplicand	0001	0000 1000	0000 0110
	3: Shift right Multiplier	0000	0000 1000	0000 0110
3	1: 0 $\Rightarrow$ No operation	0000	0000 1000	0000 0110
	2: Shift left Multiplicand	0000	0001 0000	0000 0110
	3: Shift right Multiplier	0000	0001 0000	0000 0110
4	1: 0 $\Rightarrow$ No operation	0000	0001 0000	0000 0110
	2: Shift left Multiplicand	0000	0010 0000	0000 0110
	3: Shift right Multiplier	0000	0010 0000	0000 0110

Iteration	Step	Multiplier	Multiplicand	Product
0	Initial Values	01100	00000 10100 100	000000 000000
1	1a: 0 $\Rightarrow$ No operation	01100	00000 10100 00001 01000	000000 000000
	2: Shift left M Cand	01100	00000 01000	000000 000000
	3: Shift right M Plier	00110	00001 01000	000000 000000
2	1a: 0 $\Rightarrow$ No operation	00110	00001 01000	000000 000000
	2: Shift left M Cand	00110	00010 010000	000000 000000
	3: Shift right M Plier	00011	00010 010000	000000 000000
3	1a: 1 $\Rightarrow$ Prod = Prod + M Cand	00011	00010 010000	00010 010000
	2: Shift left M Cand	00011	00101 000000 000	00010 010000
	3: Shift right M Plier	00001	00101 000000	00010 010000
4	1a: 1 $\Rightarrow$ Prod = Prod + M Cand	00001	00101 000000	00111 100000
	2: Shift left M Cand	00001	01010 000000	00111 100000
	3: Shift right M Plier	00000	01010 000000	00111 100000
5	1: 0 $\Rightarrow$ No operation	00000	01010 000000	00111 100000
	2: Shift left M Cand	00000	10100 000000	00111 100000
	3: Shift right M Plier	00000	10100 000000	00111 100000

5/

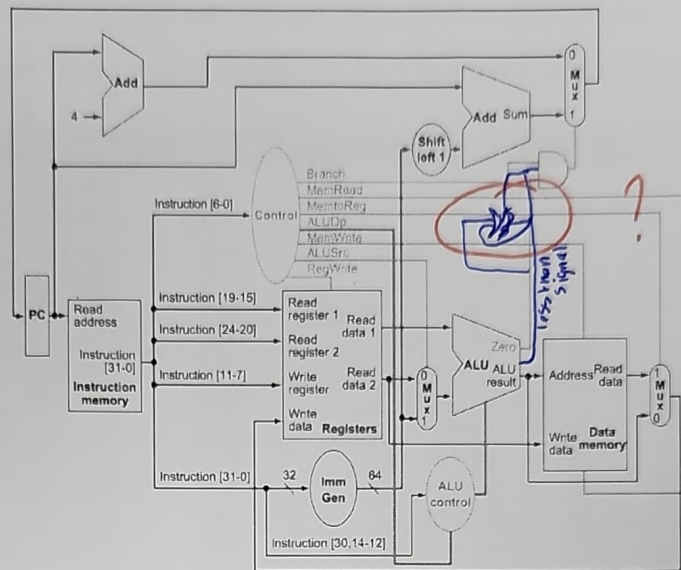
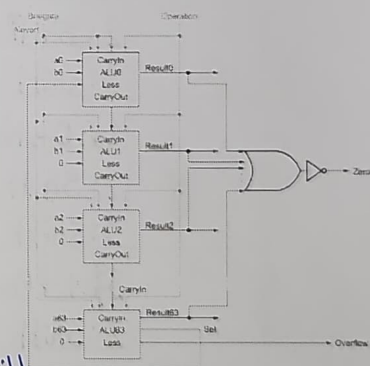
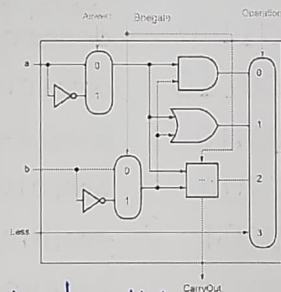


Figure 1: The single cycle datapath with control unit

Question 3 [5 points]:

For the datapath shown in Figure 1, what new datapath elements or control signals are required if another instruction bltu (branch if less than unsigned) is also supported. Draw the necessary changes required in the single cycle datapath design. (Remember we can get the 'less than' signal from the ALU)



a less than signal will be or with zero signal, the answer for this will be and with branch & go into add.

Question 4 [5 points]:

For the instructions given below, identify the datapath elements used by each instruction and the value of each control signal (ALUSrc, MemtoReg, RegWrite, MemRead, MemWrite, Branch, ALUOp1, ALUOp0)

Instruction	Datapath elements used	Control signals
add x2, x10, x11	PC, instruction memory, Registers, ALU	0,0,1,0,0,0,1,0
ld x3, 64(x2)	PC, instruction memory, Registers, Data memory, ALU	0,1,1,1,0,0,0,0

5/